ASSOCIATED PRIMES OF A MODULE WITH A PRIME-CYCLIC FILTRATION LIE AMONG THE PRIMES OF THE FILTRATION

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Allowed standard prerequisites (editorial).

- Associated primes via annihilators of elements. Used at the first step of the counted proof (line 15383). Represents an associated prime by an element whose annihilator it is.
- The submodule of elements killed by a fixed element. Used at proof lines 15388-15389. Shows that multiplying by a suitable element outside the associated prime places the witness inside the previous step of the filtration.

Lemma 0.1. *Let R be a ring, and M an R-module. Suppose there exists a filtration by R-submodules*

$$0 = M_0 \subset M_1 \subset \ldots \subset M_n = M$$

such that each quotient M_i/M_{i-1} is isomorphic to $R/\mathfrak{p}_i$ for some prime ideal $\mathfrak{p}_i$ of R. Then $Ass(M) \subset \{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\}$.

Proof. By induction on the length n of the filtration $\{M_i\}$. Pick $m \in M$ whose annihilator is a prime $\mathfrak{p}$. If $m \in M_{n-1}$ we are done by induction. If not, then m maps to a nonzero element of $M/M_{n-1} \cong R/\mathfrak{p}_n$. Hence we have $\mathfrak{p} \subset \mathfrak{p}_n$. If equality does not hold, then we can find $f \in \mathfrak{p}_n$, $f \notin \mathfrak{p}$. In this case the annihilator of fm is still $\mathfrak{p}$ and $fm \in M_{n-1}$. Thus we win by induction. □

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